



# Lakshya JEE 2.0 (2024)

## Indefinite Integration

**DPP-04**

1.  $\int \frac{dx}{\sqrt{x^2 - 2x + 3}} =$

- (1)  $\log|x-1-\sqrt{x^2-2x+3}|+c$   
 (2)  $-\log|x-1+\sqrt{x^2-2x+3}|+c$   
 (3)  $\log|x-1+\sqrt{x^2-2x+3}|+c$   
 (4)  $\log|x+1+\sqrt{x^2-2x+3}|+c$

2.  $\int \frac{5x+4}{\sqrt{x^2+2x+5}} dx$

- (1)  $-5\sqrt{x^2+2x+5} - \ln|x+1+\sqrt{x^2+2x+5}|+c$   
 (2)  $5\sqrt{x^2+2x+5} - \ln|x+1-\sqrt{x^2+2x+5}|+c$   
 (3)  $5\sqrt{x^2+2x+5} - \ln|x+1+\sqrt{x^2+2x+5}|+c$   
 (4)  $5\sqrt{x^2+2x+5} + \ln|x+1+\sqrt{x^2+2x+5}|+c$

3. If  $\int \frac{\cos\theta \cdot d\theta}{5+7\sin\theta-2\cos^2\theta} d\theta = A \log|B(\theta)|+c$ , then  $\frac{B(\theta)}{A}$  can be:

- (1)  $\frac{2\sin\theta+1}{5(\sin\theta+3)}$  (2)  $\frac{2\sin\theta+1}{\sin\theta+3}$   
 (3)  $\frac{5(2\sin\theta+1)}{\sin\theta+3}$  (4)  $\frac{5\sin\theta}{2\sin\theta+1}$

4.  $\int \sqrt{\cot\theta} d\theta$

5. If  $\int \frac{2e^x + 3e^{-x}}{4e^x + 7e^{-x}} dx = \frac{1}{14} (ux + v \ln|4e^x + 7e^{-x}|) + c$

Then  $u+v$  is equal to:

6.  $\int \frac{x^2+1}{x^4-x^2+1} dx$  is equal to

- (1)  $\tan^{-1}\left(\frac{x^2-1}{x}\right)+c$   
 (2)  $\tan^{-1}\left(\frac{x^2+1}{2x}\right)+c$   
 (3)  $-\tan^{-1}\left(\frac{x^2-1}{x}\right)+c$   
 (4)  $\tan^{-1}\left(\frac{x^2-1}{2x}\right)+c$

7.  $\int \frac{x^2+4}{x^4+16} dx = \frac{1}{k} \tan^{-1}\left(\frac{x^2-4}{kx}\right) + c$

- (1)  $\sqrt{2}$  (2)  $4\sqrt{2}$   
 (3)  $2\sqrt{2}$  (4)  $2$

8.  $\int \frac{e^{3x} + e^x}{e^{4x} - e^{2x} + 1} dx =$

- (1)  $\frac{1}{4} \log(e^{4x} - e^{2x} + 1) + c$   
 (2)  $\tan^{-1}(e^x - e^{-x}) + c$   
 (3)  $\tan^{-1}(e^x + e^{-x}) + c$   
 (4)  $\tan^{-1}(e^{-x} - e^x) + c$

9. For real numbers  $\alpha, \beta, \gamma$  and  $\delta$  is

$$\int \frac{(x^2 - 1) + \tan^{-1}\left(\frac{x^2 + 1}{x}\right)}{(x^4 + 3x^2 + 1) \tan^{-1}\left(\frac{x^2 + 1}{x}\right)} dx$$

$$= \alpha \cdot \ln\left(\tan^{-1}\left(\frac{x^2 + 1}{x}\right)\right)$$

$$+ \beta \tan\left(\frac{\gamma(x^2 - 1)}{x}\right) + \delta \cdot \tan^{-1}\left(\frac{x^2 + 1}{x}\right) + c,$$

where  $c$  is an arbitrary constant, then the value of  $10(\alpha + \beta\gamma + \delta) =$

10. Suppose

$$A = \int \frac{dx}{x^2 + 6x + 25} \text{ and } B = \int \frac{dx}{x^2 - 6x - 27}$$

$$\text{If } 12(A + B) = \lambda \cdot \tan^{-1}\left(\frac{x+3}{4}\right) + \mu \cdot \ln\left|\frac{x-9}{x+3}\right| + C,$$

then the value of  $(\lambda + \mu)$  is:

- (1) 3
- (2) 4
- (3) 5
- (4) 6

**Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.**

## Answer Key

1. (3)

2. (3)

3. (3)

4. 
$$-\left(\frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{\cot \theta - 1}{\sqrt{2} \cot \theta}\right) + \frac{1}{2\sqrt{2}} \log \left| \frac{\cot \theta - \sqrt{2} \cot \theta + 1}{\cot \theta + \sqrt{2} \cot \theta + 1} \right| \right) + C$$

5. (7)

6. (1)

7. (3)

8. (2)

9. (6)

10. (2)

